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Eigen Values & Eigen vectors :

$$A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$$

Characteristic equation: $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 2 \\ 5 & 4-\lambda \end{vmatrix} = 0$$

$$(4-\lambda)(1-\lambda) - 10 = 0$$

$$4 - 4\lambda - \lambda + \lambda^2 - 10 = 0$$

$$+\lambda^2 - 5\lambda - 6 = 0$$

$\therefore$ Eigen values: $\lambda_1 = 6$, $\lambda_2 = -1$

Eigen vectors:

$\lambda_1 = 6$

$$[A - 6I]\, v_1 = 0$$

$$\begin{bmatrix} 1-6 & 2 \\ 5 & 4-6 \end{bmatrix} v_1 = \begin{bmatrix} -5 & 2 \\ 5 & -2 \end{bmatrix} v_1 = 0$$

$-5x + 2y = 0$ $\quad y = \frac{5}{2}x$

$5x - 2y = 0$

if $x = 2$

$y = 5$

$$\boxed{v_1 = \begin{bmatrix} 2 \\ 5 \end{bmatrix}}$$

$\lambda_2 = -1$ $\quad [A - (-1)I]\, v_2 = 0$

$$\begin{bmatrix} 2 & 2 \\ 5 & 5 \end{bmatrix} v_2 = 0 \quad \begin{matrix} 2x + 2y = 0 \\ 5x + 5y = 0 \end{matrix}$$

$x + y = 0$

$x = -y$

$x = 1$, $y = -1$

$$\boxed{v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}}$$

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→ PCA :

It is a statistical technique used to reduce the no. of variables (dimensions) in a dataset.

Key concept is based on:
- covarience matrix
- Eigen values & Eigenvectors

covarience ; $C = \frac{1}{n-1} \Sigma (x_i - \bar{x}_i)(x_j - \bar{x}_j)$

Q.

F	Ex1	Ex2	Ex3	Ex4
x_1	4	8	13	7
x_2	11	4	5	14

Sol:- Mean,

$\bar{x}_1 = 8$

$\bar{x}_2 = 8.5$

covariance matrix

$$S = \begin{bmatrix} cov(x_1, x_1) & cov(x_1, x_2) \\ cov(x_2, x_1) & cov(x_2, x_2) \end{bmatrix}$$

$N = 4$

$$cov(x_1, x_1) = \frac{1}{N-1} \Sigma (x_1 - \bar{x}_1)(x_1 - \bar{x}_1)$$

$$= \frac{1}{3}\left[(4-8)^2 + (8-8)^2 + (13-8)^2 + (7-8)^2\right]$$

$$= 14$$

$$cov(x_1, x_2) = \frac{1}{3}\left[(4-8)(11-8.5) + (8-8)(4-8.5) + (13-8)(5-8.5) + (7-8)(14.8-5\right.$$

$$= -11$$

$\text{cov}(x_2, x_1) = -11$

$\text{cov}(x_2, x_2) = 23$

$$S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

Eigen values,

$$|S - \lambda I| = 0$$

$$\begin{vmatrix} 14-\lambda & -11 \\ -11 & 23-\lambda \end{vmatrix} = 0$$

$$(14-\lambda)(23-\lambda) - (-11)(-11) =$$
$$332 - 14\lambda - 23\lambda + \lambda^2 - 121$$
$$= \lambda^2 - 37 + 201$$

$$\lambda_1 = 30.38$$
$$\lambda_2 = 6.61$$

Eigen vectors

$\lambda_1 = 30.38$

$$[S - \lambda I]\, v_1 = 0 \qquad \begin{bmatrix} 14-\lambda & -11 \\ -11 & 23-\lambda \end{bmatrix} v = 0$$

$$(14-\lambda) v_1 - 11 v_2 = 0$$
$$11 v_1 + (23-\lambda) v_2 = 0$$

$$\cancel{-16.38 v_1 - 11 v_2 = 0}$$

$$14-\lambda \, v_1 = 11 \, v_2$$

$$\frac{v_1}{11} = \frac{v_2}{14-\lambda} = t$$

$$v_1 = 11t$$
$$v_2 = (14-\lambda)t$$

$$v = \begin{bmatrix} 11 \\ 14-\lambda \end{bmatrix}$$

*consider largest Eigen value
$\lambda_1 = 30.38$

find unit eigen vector,

$$|v_1| = \sqrt{11^2 + (14-\lambda_1)^2}$$

$$= \sqrt{11^2 + (14-30.38)^2} = 19.73$$

$$e_1 = \begin{bmatrix} 11/19.73 \\ 14-30.38/19.73 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5574 \\ -0.8303 \end{bmatrix}$$

$e_1^T = \cancel{[0.5574}$

$e_1^T = [0.5574 \quad -0.8303]$

$$e_2 = \begin{bmatrix} 0.8303 \\ 0.5574 \end{bmatrix}$$

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Principal component

$$e_1^T \begin{bmatrix} x_{1k} - \bar{x}_1 \\ x_{2k} - \bar{x}_2 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5574 & -0.8303 \end{bmatrix} \begin{bmatrix} x_{11} - x_1 \\ x_{21} - x_2 \end{bmatrix}$$

$$= 0.5574 (x_{11} - \bar{x}_1) - 0.8303 (x_{21} - x_2)$$

$$= 0.5574 (4 - 8) - 0.8303 (11 - 8.5)$$

$$= -4.30535$$

Quick Work

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→ SVD :

SVD is a matrix factorization technique that breaks any matrix into simpler matrices.

$$A = U \Sigma V^T$$

Q. $A = \begin{bmatrix} 1 & 1 \\ 7 & 7 \end{bmatrix}$

Sol: Find A^T,

$$A^T = \begin{bmatrix} 1 & 7 \\ 1 & 7 \end{bmatrix}$$

Calculate $A \times A^T$

$$= \begin{bmatrix} 2 & 14 \\ 14 & 98 \end{bmatrix}$$

$$A^T \times A = \begin{bmatrix} 50 & 50 \\ 50 & 50 \end{bmatrix}$$

Find, $U \Rightarrow$ Eigen Values & eigen vectors of AA^T

$$|AA^T - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & 14 \\ 14 & 98-\lambda \end{vmatrix} = 0$$

$$\lambda^2 - 100\lambda = 0$$

$$\lambda = 0, \lambda = 100$$

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Eigen vectors

$\lambda = 0$

$$\begin{bmatrix} 2 & 14 \\ 14 & 98 \end{bmatrix} V_1 = 0$$

$$2x + 14y = 0$$
$$14x + 98y = 0$$

$$14y = -2x$$
$$7y = -x \Rightarrow x = -7y$$
$$x = 1 \quad y = -7 \qquad y = 1 \quad x = -7$$

$$\Rightarrow V_1 \begin{bmatrix} -7 \\ 1 \end{bmatrix}$$

$$\|V_1\| = \sqrt{(-7)^2 + (1)^2} = \sqrt{50} = 7.07$$

Unit eigen vectors:

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -7/7.07 \\ 1/7.07 \end{bmatrix} = \begin{bmatrix} -0.99 \\ 0.14 \end{bmatrix}$$

$\lambda = 100$

$$\begin{bmatrix} 2 & 14 \\ 14 & 98 \end{bmatrix} V_2 = 0$$

$$V2 = \begin{bmatrix} 0.14 \\ 0.99 \end{bmatrix}$$

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(*) arrang Evec in dec order

$$U = \begin{bmatrix} 0.14 & -0.99 \\ 0.99 & 0.14 \end{bmatrix}$$

Find V^T

Eigen values & vectors of $A^T X A$

$$\begin{bmatrix} 50-\lambda & 50 \\ 50 & 50-\lambda \end{bmatrix} = 0$$

$$\lambda^2 - 100\lambda = 0$$

$$\lambda = 0, \lambda = 100$$

$$V_1 = \begin{bmatrix} -0.70 \\ 0.70 \end{bmatrix}$$

$$V_2 = \begin{bmatrix} 0.70 \\ 0.70 \end{bmatrix}$$

$$V^T = \begin{bmatrix} 0.70 & -0.70 \\ 0.70 & 0.70 \end{bmatrix}$$

find Σ

in matrix all elements are 0 except the diagonals that are $\sqrt{\text{eigen values}}$..

$$\Sigma = \begin{bmatrix} \sqrt{100} & 0 \\ 0 & \sqrt{0} \end{bmatrix} = \begin{bmatrix} 10 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 7 & 7 \end{bmatrix} = U \Sigma V^T = \begin{bmatrix} 0.14 & -0.99 \\ 0.99 & 0.14 \end{bmatrix} \times \begin{bmatrix} 10 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\times \begin{bmatrix} 0.70 & -0.70 \\ 0.70 & 0.70 \end{bmatrix}$$

$$= \begin{bmatrix} 0.98 & -0.98 \\ 6.93 & -6.93 \end{bmatrix}$$

Quick Work

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Chi - Square test :

Q. chi- square test is a statistical test used to compare observed frequencies with expected frequencies.

used for :

- Testing independence of two variables.
- Testing goodness of fit.

$$\boxed{\chi^2 = \sum \frac{(O-E)^2}{E}}$$

Q. In a survey of 60 students:

Gender	Tea	coefee	total
Boys	20	10	30
girls	10	20	30
total	30	30	60

Test for independence :-

Hypothesis :

Step 1: H_0 : Gender & drink are independent

H_1 : They are dependent

Step 2: significance

$\alpha = 0.05$

Step 3: dof

$df = (r-1)(c-1)$

r - row

c - column

$\Rightarrow df = (2-1)(2-1) = 1$

Step 4: critical value

$$\chi^2_{(0.05,1)} = 3.84$$

Step 5: Expected frequencies

$$E = \frac{(\text{Row total}) \cdot (\text{Column total})}{N\ (\text{total})}$$

Boys - Tea :

$$E = \frac{30 \times 30}{60} = 15$$

Boys - coffee : $E = \frac{30 \times 30}{60} = 15$

Girls - Tea : $E = \frac{30 \times 30}{60} = 15$

Girls - Cofee $E = \frac{30 \times 30}{60} = 15$

Step 6: $\sum \frac{(O - E)^2}{E}$

Boys - Tea : $O = 20,\ E = 15$

$$\frac{(20-15)^2}{E} = \frac{25}{15} = 1.67$$

Boys - coffe: $\frac{25}{15} = 1.67$

Girls - Tea : $\Rightarrow 1.67$

Girls - coffee $\Rightarrow 1.67$

$$\chi^2 = 1.67 + 1.67 + 1.67 + 1.67$$

$$\chi^2 = 6.68$$

$$\chi^2_{cal} > \chi^2_{tab\ (crit)} \rightarrow \text{Reject } H_0$$

$$\chi^2_{cal} < \chi^2_{tab} \rightarrow \text{accept } H_0$$

$$6.68 > 3.84$$

$\therefore$ H_0 is Rejected.

Test for Goodness

Drink	observed
Tea	30
coffee	30
total	60

Hypothesis:

H_0: distribution is equal (50-50)
H_1: distribution is unequal

Expected frequencies

$$E = \frac{60}{2} = 30$$

$$\sum \frac{(O-E)^2}{E} \Rightarrow \frac{(30-30)^2}{30} = 0$$

Tea

Coeffee $\frac{30-30}{30} = 0$

$\chi^2 = 0 + 0 = 0$

dof = $K-1 = 2-1 = 1$

$\chi_{(0.05,1)} = 3.84$

$\chi_{cal} < \chi_{tab}$

Ho is accepted.